

UC MATH 600 MATHEMATICS FOR COMPUTER SCIENCE · MODULE 13 · LECTURE NOTES

Claiming You Proved Something

Finding your own gaps, and knowing you are ready.

MODULE OUTCOMES

- State what your proof actually establishes.
- Find gaps in your own arguments.
- Write a proof somebody else can check.
- Assess honestly whether you are ready for Semester 1.
- State the program's closing rule in this subject.

1 What a proof establishes

A PROOF ESTABLISHES THE STATED CLAIM UNDER THE STATED HYPOTHESES, AND NOTHING BROADER

Which hypotheses — including the ones you used without noticing: that n is positive, that the graph is simple (Module 07 §1), that the list is non-empty, that the set is finite (Module 04 §1's empty-case trap).

What exactly was proved — 'for all n ' and 'for all sufficiently large n ' are different claims, and both are common; so are 'there exists' and 'we can construct'.

And what was assumed from elsewhere — a theorem you cited is a hypothesis you did not check, and if its conditions do not hold in your setting then your proof does not either.

So the honest form is: 'under these assumptions, this claim, by this argument' — three things, and the first is the one people omit, usually because the assumptions felt too obvious to mention, which is exactly when they are worth mentioning.

2 Finding your own gaps

- 1 EVERY "OBVIOUSLY", "CLEARLY", "IT FOLLOWS THAT" is a step you did not write down. Write it. (Module 01 section 1)
- 2 EVERY HYPOTHESIS: where was it used? if nowhere, then either the claim is stronger than you stated it or the proof is wrong.
- 3 EVERY INDUCTION: point at the exact line where the inductive hypothesis was used, and say which predecessors it needed. (Module 03 section 1)
- 4 EVERY EDGE CASE: $n = 0$, $n = 1$, the empty set, the single-element list, the disconnected graph. Does the argument still run?
- 5 EVERY QUANTIFIER: is it "for all x there is a y " or "there is a y for all x "? (Module 02 section 1)
- 6 AND: would somebody who actively wants to disbelieve this accept every single line?

RUN THIS ON A PROOF A WEEK OLD, NOT A FRESH ONE.

BECAUSE YOU CANNOT SEE THE GAP IN AN ARGUMENT YOU JUST BUILT

While you are writing, you are holding the entire structure in your head — so the missing step is present in your mind and absent from the page, and you read it as though it were there, every time, no matter how carefully you look.

A week later you are the sceptical reader (Module 01 §1) — which is the reader the proof was supposed to be written for in the first place, and the only one whose acceptance means anything.

And reading it aloud catches a surprising amount, because speech will not let you skim the connectives: 'therefore' and 'it follows that' have to be said, and saying them makes you notice whether they are earned.

Which is CSCE 671 Module 01's designer blindness arriving in mathematics — you cannot experience your own work as unfamiliar — and every remedy is some way of manufacturing distance: time, reading aloud, or another person. Project 2's self-audit is exactly this checklist, and it reliably finds something.

3 Are you ready?

- Can you prove something by induction with all three of Module 03's requirements visible — base case computed, hypothesis named, use pointed at — **without looking anything up?**
- Can you prove $3n^2 + 7n = O(n^2)$ by exhibiting a c and an n_0 ? (Module 09 §2.)
- Can you solve $T(n) = 2T(n/2) + n$ and say which case of the master theorem it is, and why? (Module 10.)
- Can you do the 99% test calculation by counting a million people, and explain why the answer is 1%? (Module 11 §3.)
- And can you compute an expected count by indicator variables, and say why dependence does not matter? (Module 12 §2.) **Five questions. If you can answer all five without reference, start Semester 1; if not, the ones you failed name exactly the modules to revisit, which is a far more useful outcome than a score.**

4 The rule

STATE WHAT YOU PROVED, STATE WHAT YOU ASSUMED, AND NEVER CLAIM MORE THAN YOU ESTABLISHED

This is the rule that every one of the thirty-six courses in this program closes on, and it begins here, in its original form — because a proof is where the distinction between what is established and what is merely believed is sharpest and most checkable.

In CSCE 629 it becomes a stated complexity with its machine model named; in CSCE 628 a five-part methods section; in CSCE 640 a problem, a baseline, and a machine; in CSCE 717 a solution concept, an agent model, and an impossibility.

And they are all this one instruction: a claim is a claim about an argument, and the argument is the part somebody else can check — which is why the argument is what gets written down.

Which is also why the proof portfolio keeps the wrong versions (Project 1's requirement) — a record of what you believed and why it failed is worth more than a record of what turned out to be true, because the second teaches nothing about how you go wrong.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Velleman — How to Prove It, the final chapters	§§1 and 2 — writing proofs for a reader, which is the skill this module is about. Library copy.
Lamport — How to Write a 21st Century Proof (free)	§2 — a structured proof format designed so that gaps are visible, by somebody who found errors in published work.
Lehman, Leighton & Meyer, the false proofs (free)	§2 — worked examples of arguments that look right and are not, which is the best practice material there is.
Pólya — How to Solve It, the looking-back chapter	§2 — the step after finding a proof, which is where the learning is. Library copy.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State the three parts** of an honest claim for one of your proofs.
- 2 **List the hypotheses you used without noticing** in three proofs.
- 3 **Find a cited theorem** whose conditions you did not check.
- 4 **Run the six-point audit** on a proof you wrote last month.
- 5 **Read one of your proofs aloud** and note what you stumble on.
- 6 **Swap proofs with somebody** and audit each other's.
- 7 **Find three published false proofs** and locate the gap in each.
- 8 **Take the five readiness questions** without reference material.
- 9 **Name the modules** your failures point at, and spend a week there.
- 10 **Project 2 is now due.** Submit the full thirty-proof portfolio, the wrong versions with their errors named, the three recurrences solved three ways, and the self-audit with at least two gaps found in your own earlier work.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What does a proof establish, in three parts?
- 2 Why is a cited theorem a hypothesis?
- 3 Give the six-point audit.
- 4 Why run it on an old proof rather than a fresh one?

- 5 Why can you not see your own gaps?
- 6 What two remedies manufacture distance?
- 7 Give the five readiness questions.
- 8 What should you do with the ones you fail?
- 9 State the program's closing rule.
- 10 Why does the portfolio keep the wrong versions?